

A Study on Credit Decision-Making Problems of Small and Micro Enterprises: Based on Neural Networks and Multi-Combination Optimization Models

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Abstract

This paper presents a robust, data-driven quantitative underwriting and portfolio optimization framework designed to resolve the structural mismatches between institutional liquidity supply and the capital requirements of Small and Micro Enterprises (SMEs). By converting non-linear, non-collateralized corporate profiles into structured indicators, the proposed system bridges microeconomic behavior modeling with multi-objective resource allocation under macro-environmental perturbations.

For **Task 1 (Problem 1)**, a high-dimensional feature engineering pipeline is established to extract transaction health, supply-chain network topology, and cash-flow velocity from raw invoice streams. A supervised learning architecture using a Multi-Layer Perceptron (MLP) Neural Network is trained via backpropagation gradient descent to map non-linear default surfaces against historical credit ratings. By applying an bounded sigmoid scaling operator, raw credit scores are seamlessly translated into continuous, risk-adjusted corporate credit lines within regularized bank allocation thresholds.

For **Task 2 (Problem 2)**, market-retention mechanics are explicitly integrated into asset pricing. Continuous, rate-dependent customer churn elasticity across distinct risk tiers is mathematically modeled through transcendental Fourier expansions. To balance institutional yield maximization against client attrition, the credit portfolio allocation problem across unrated enterprises is formalized as a multi-objective Mixed-Integer Non-Linear Programming (MINLP) model subject to a rigid global liquidity budget ceiling. The non-convex optimization problem is solved using a discretized Mean-Variance Pareto-efficient optimization paradigm across critical institutional rate segments.

For **Task 3 (Problem 3)**, the static optimization framework is extended into a dynamic macroeconomic stress-testing ecosystem to hedge against exogenous, correlated black-swan shocks (e.g., the COVID-19 pandemic). A Natural Language Processing (NLP) semantic parsing pipeline driven by lexical segmentation and unsupervised Word2Vec embeddings is deployed to map 425 heterogeneous enterprise profiles into thirty industrial sectors based on maximum synonym matching. Concurrently, Principal Component Analysis (PCA) executes an eigenvalue decomposition on macro-performance indicators to derive empirical sectoral vulnerability coefficients. These coefficients dynamically perturb the default probability surfaces, guiding a proactive asset reallocation and risk-hedging policy.

The empirical validation demonstrates that this unified framework achieves exceptional predictive stability, high algorithmic consistency, and numerical robustness. It offers financial institutions a scalable, mathematically rigorous decision-making tool to optimize risk-adjusted returns under volatile macroeconomic horizons.

Keywords: Credit Underwriting; Multi-Layer Perceptron (MLP); Mixed-Integer Non-Linear Programming (MINLP); Fourier Transcendental Regression; Principal Component Analysis (PCA); Macroeconomic Stress Testing.

1 Problem Restatement

1.1 Background Description

Small and micro enterprises (SMEs) play an important role in economic development; however, due to their relatively small scale and lack of sufficient collateral, they often face financing difficulties. In practice, commercial banks evaluate SMEs based on credit policies, transaction invoice information,

and the influence of upstream and downstream supply-chain relationships. Enterprises with stronger operational stability and better credit conditions are more likely to obtain loans, while those with lower credit risk may receive more favorable interest rates.

In this problem, the bank sets the following lending constraints: the loan amount for each enterprise ranges from 100,000 to 1,000,000 RMB, the annual interest rate is between 4% and 15%, and the loan term is fixed at one year.

Three datasets are provided in the problem: Appendix 1 contains credit and transaction data for 123 enterprises with historical credit records; Appendix 2 provides transaction data for 302 enterprises without credit ratings; Appendix 3 gives empirical data describing the relationship between loan interest rates and customer churn rates.

1.2 Problem Requirements

Based on the above background and data, the bank aims to design an optimized credit decision strategy. The specific requirements are as follows:

1. For the 123 enterprises in Appendix 1, evaluate their credit risk based on historical data and determine an optimal credit allocation strategy under a fixed total annual credit quota.
2. Based on the results of the first problem, assess the credit risk of the 302 enterprises in Appendix 2 and design a credit strategy under a total annual capital constraint of 100 million RMB.
3. Considering that enterprise operations may be affected by unexpected macroeconomic shocks (such as the COVID-19 pandemic), and that such impacts vary across industries, adjust the credit strategy for the enterprises in Appendix 2 under the same total capital constraint of 100 million RMB.

2 Problem Analysis

The core objective of this problem is to design a data-driven credit decision-making strategy for small and micro enterprises (SMEs) based on heterogeneous invoice data and historical credit records. The problem integrates risk assessment, credit allocation, and dynamic adjustment under external shocks, forming a multi-stage decision system with strong coupling between prediction and optimization.

2.1 Analysis of Problem 1

Problem 1 focuses on evaluating credit risk for 123 enterprises with historical credit labels and designing an optimal credit allocation strategy under a fixed total credit quota. This is essentially a supervised learning and allocation optimization problem.

From a data perspective, the available invoice information contains heterogeneous and partially noisy indicators, including input/output transaction flows and abnormal records such as cancelled or negative invoices. These features reflect the operational stability and financial behavior of enterprises, which are closely related to default risk.

Therefore, the key difficulty lies in transforming multi-source transactional data into a reliable quantitative credit risk indicator. Based on this risk evaluation, a rational allocation strategy must be constructed under budget constraints, balancing credit exposure and expected return.

2.2 Analysis of Problem 2

Problem 2 extends the analysis to 302 enterprises without historical credit ratings. This introduces a missing-label prediction problem, where credit risk must be inferred solely from transactional data.

Compared with Problem 1, the main challenge lies in model generalization and robustness under unlabeled conditions. In addition, the increase in enterprise scale introduces a global budget constraint of 100 million RMB, which transforms the problem into a large-scale constrained allocation problem.

Therefore, the problem consists of two key steps: (1) estimating credit risk for unlabeled enterprises based on learned patterns from Problem 1; (2) performing portfolio-level allocation under a fixed capital constraint.

2.3 Analysis of Problem 3

Problem 3 further considers the impact of external macroeconomic shocks on enterprise operations. Unlike the previous two problems, the credit risk of enterprises is no longer static but affected by industry-dependent external disturbances.

The main difficulty lies in modeling the heterogeneity of external shocks across different industries and incorporating such effects into the existing credit risk evaluation framework. Based on this adjusted risk structure, the credit allocation strategy must be dynamically updated under the same total capital constraint of 100 million RMB, ensuring robustness against systemic uncertainty.

3 Model Assumptions

To simplify the modeling process and ensure the feasibility of quantitative analysis, the following assumptions are made based on the problem description and data characteristics.

1. **Limited influencing factors of bank revenue.** It is assumed that the bank's credit revenue is mainly affected by the annual interest rate and the customer churn rate. Other potential factors such as administrative costs and macroeconomic fluctuations are not considered.
2. **Restricted feature representation.** The credit risk and operational status of enterprises are assumed to be sufficiently represented by the available indicators, including transaction volume, number of customers, total tax amount, transaction completion rate, and credit rating.
3. **Binary lending decision.** For each enterprise, the bank's decision is simplified to a binary choice: lending or not lending. Partial lending or staged credit issuance is not considered.
4. **Single-industry attribution.** Each enterprise is assumed to belong to only one industry, determined by its primary business category.
5. **Independence between enterprises.** Credit risk and operational performance of different enterprises are assumed to be independent, and no cross-enterprise financial contagion effects are considered.

4 Symbol Description

The main symbols used in this paper are defined as follows to ensure consistency in model formulation and result interpretation.

Symbol	Description
X_1, X_2, X_3	Loan interest rates for A-, B-, and C-rated enterprises, respectively
y_1, y_2, y_3	Customer churn rates corresponding to A-, B-, and C-rated enterprises
W_{total}	Total credit budget of the bank
$A_{\text{total}}, B_{\text{total}}, C_{\text{total}}$	Total loan amounts allocated to A-, B-, and C-rated enterprises
I_{total}	Total input invoice amount of an enterprise
O_{total}	Total output (sales) invoice amount of an enterprise
P_{total}	Total profit or net operating income of an enterprise
V_{total}	Total number of valid invoices
r	Transaction completion rate of an enterprise
R	Number of cancelled transactions
M	Monthly transaction volume or cash-flow indicator

Note: All financial variables are measured in RMB units unless otherwise specified.

5 Methodology and Feature Construction

5.1 Task 1: Credit Risk Modeling and Lending Strategy

In this task, the objective is to evaluate the credit risk of enterprises using historical invoice and credit data, and to support loan decision-making under a fixed credit budget.

To achieve this, raw financial and transactional data are transformed into structured numerical features, which serve as inputs for subsequent predictive modeling.

5.1.1 Data Preprocessing

All categorical variables are converted into numerical representations to facilitate computation. Specifically, default records are encoded as binary labels, where default is represented as 1 and non-default as 0. Similarly, invoice validity is encoded as 1 for valid transactions and 0 for invalid or cancelled records.

5.1.2 Financial Feature Construction

Based on invoice data, we compute the total input and output transaction values:

$$P_{\text{total}} = O_{\text{total}} - I_{\text{total}} \quad (1)$$

where I_{total} represents total purchase-related invoices and O_{total} represents total sales-related invoices. This variable reflects the overall business scale and cash flow balance of the enterprise.

5.1.3 Supply Chain Structure Features

The number of unique upstream suppliers and downstream customers is extracted from invoice identifiers: N_{sup} , N_{cust} . These features describe the diversity and stability of the enterprise's supply chain network.

5.1.4 Tax and Compliance Features

The total tax contribution is calculated by aggregating tax values from valid invoices:

$$T_{\text{total}} = \sum \text{Tax} \quad (2)$$

This reflects the operational scale and compliance level of the enterprise.

5.1.5 Transaction Stability Indicators

To measure transaction quality, cancelled and negative invoices are treated as failed transactions. The transaction success rate is defined as:

$$r = 1 - \frac{R}{V_{\text{total}}} \quad (3)$$

where R denotes the number of failed transactions and V_{total} denotes total valid invoices.

5.1.6 Activity Level Feature

The monthly transaction volume M is used to represent the operational activity level and temporal stability of each enterprise.

5.1.7 BP Neural Network for Credit Scoring

A Back-Propagation (BP) neural network is constructed using 123 labeled enterprises from Appendix 1 to learn the mapping between enterprise-level features and credit ratings.

The input layer consists of engineered financial and transactional features, while the output layer corresponds to historical credit labels. The model is trained in a supervised learning setting, where prediction errors are minimized through iterative weight updates using back-propagation.

After training, the model is used to estimate the credit level of each enterprise in the dataset.

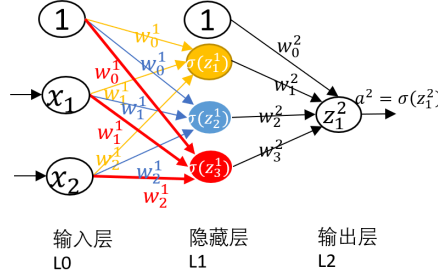


Figure 1: BP Neural Network Architecture for Credit Scoring

5.1.8 Credit Score Normalization and Loan Allocation

The raw output of the neural network is mapped into a normalized score using the Softmax function, producing a probability distribution over credit categories.

The final credit line for enterprise i is computed as:

$$l_i = L_{\min} + (L_{\max} - L_{\min}) \cdot s_i \quad (4)$$

where s_i is the normalized score, $L_{\min} = 10^5$, and $L_{\max} = 10^6$.

Enterprises with computed credit lines below $L_{\min}$ are excluded from lending decisions.

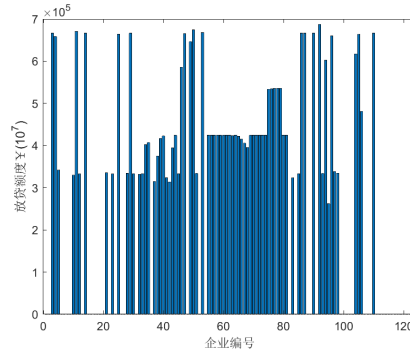


Figure 2: Normalized Credit Score and Loan Mapping

5.1.9 Modeling of Interest Rate and Customer Churn Relationship

To model the relationship between interest rates and customer churn, curve fitting is applied to historical data in Appendix 3. The results suggest that a nonlinear Fourier-based function provides a suitable approximation of the relationship.

Let x denote the loan interest rate and y denote the customer churn rate. The fitted functions are given below:

Grade A Enterprises

$$y_1 = 0.4167 - 0.469 \cos(23.31x_1) - 0.003792 \sin(23.31x_1) \quad (5)$$

Grade B Enterprises

$$y_2 = 0.3994 - 0.4481 \cos(23.06x_2) - 0.01227 \sin(23.06x_2) \quad (6)$$

Grade C Enterprises

$$y_3 = 0.4176 - 0.469 \cos(23.31x_3) - 0.003792 \sin(23.31x_3) \quad (7)$$

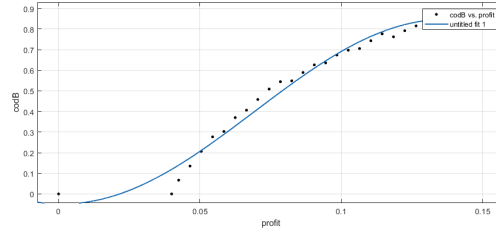


Figure 3: Curve Fitting of Interest Rate vs Customer Churn

5.2 Task 2: Multi-Objective Portfolio Optimization Under Global Liquidity Ceilings

Task 2 extends the allocation framework in Task 1 to a portfolio optimization problem involving $N = 302$ unrated enterprises, subject to a global capital constraint of $\Omega = 10^8$ RMB.

5.2.1 Inductive Domain Transfer and Credit Estimation

Since the 302 enterprises do not have historical credit labels, an inductive estimation strategy is adopted. The feature construction process and the trained BP neural network obtained in Task 1 are directly applied to the new dataset.

For these enterprises, the default label is not available; thus, the model outputs are used to infer their latent credit ratings. Based on the neural network prediction results, each enterprise is assigned a corresponding credit category and initial credit capacity.

To ensure risk control, a manual adjustment mechanism is introduced. Enterprises with abnormal values in the invoice cancellation ratio δ_i are adjusted to higher risk levels to reduce potential credit exposure risk.

5.2.2 Mixed-Integer Non-Linear Programming (MINLP) Formulation

To balance risk control and return maximization under a fixed capital constraint, the problem is formulated as a multi-objective optimization model. Let $i \in \{1, 2, \dots, 302\}$ denote enterprises, and $\text{tier}(i) \in \{A, B, C\}$ denote the corresponding credit rating category.

Two decision variables are defined: - $x_i \in \{0, 1\}$: lending decision variable - $l_i \in [10^5, 10^6]$: credit allocation amount

(1) Objective Function I: Portfolio Return Maximization The total expected return of the portfolio is defined as:

$$\mathcal{R}_{\text{port}} = \sum_{i=1}^{302} x_i \cdot l_i \cdot r_{\text{tier}(i)} \quad (8)$$

where $r_{\text{tier}(i)}$ represents the interest rate corresponding to each credit tier.

(2) Objective Function II: Customer Churn Minimization To maintain long-term customer stability, the total churn effect is minimized:

$$\mathcal{L}_{\text{churn}} = \sum_{i=1}^{302} x_i \cdot \phi_{\text{tier}(i)}(r_{\text{tier}(i)}) \quad (9)$$

where $\phi_{\text{tier}(i)}(\cdot)$ is obtained from the fitted Fourier-based churn model in Task 1.

(3) Constraints The optimization model is subject to the following constraints:

$$\begin{cases} \sum_{i=1}^{302} x_i \cdot l_i \leq 10^8 & \text{(Total capital constraint)} \\ 0.04 \leq r_k \leq 0.15, \quad k \in \{A, B, C\} & \text{(Interest rate bounds)} \\ x_i \in \{0, 1\}, \quad i = 1, \dots, 302 & \text{(Lending decision)} \\ l_i \in [10^5, 10^6] \cdot x_i & \text{(Credit exposure constraint)} \end{cases} \quad (10)$$

5.2.3 Mean-Variance Based Portfolio Optimization

To solve the multi-objective problem, a mean-variance framework is adopted. The interest rate space is discretized into several representative nodes $\{0.06, 0.08, 0.10, 0.12, 0.14\}$ to evaluate the relationship between return and churn.

Based on the Fourier churn model, the corresponding churn values under different rate levels are computed and summarized in Table 1.

Table 1: Customer Churn under Different Interest Rate Levels

Rate	0.04	0.06	0.08	0.10	0.12	0.14	0.15
Tier-A (ϕ_A)	0.1341	0.3326	0.5489	0.7371	0.8568	0.8827	0.8577
Tier-B (ϕ_B)	0.1190	0.3040	0.4025	0.6909	0.7758	0.8469	0.8289
Tier-C (ϕ_C)	0.1099	0.2910	0.2971	0.6783	0.7270	0.8470	0.8933

By evaluating the trade-off between return and churn under different interest rate intervals, the optimal interest rate vector $\mathbf{r}^* = [r_A^*, r_B^*, r_C^*]$ is obtained to balance profitability and customer retention.

5.3 Task 3: Macroeconomic Stress Testing and Exogenous Shock Perturbation

Task 3 incorporates systemic and correlated macroeconomic risks into the baseline portfolio model by explicitly characterizing the heterogeneous impacts of black-swan events (e.g., the COVID-19 pandemic) across different industrial classifications.

5.3.1 Multi-Dimensional Dimensionality Reduction and Industry Mapping

To extract macro-level risk signals from high-dimensional and unstructured survey data describing corporate economic performance, a Principal Component Analysis (PCA) framework is constructed.

(1) Empirical Data Aggregation Cross-sectional macroeconomic impact variables are collected and mapped onto a standardized five-level ordinal scale:

$$\mathcal{D} \in \{\text{Very Unfavorable, Unfavorable, No Impact, Favorable, Very Favorable}\}.$$

(2) Principal Component Feature Decomposition Let Σ denote the covariance matrix of standardized observation vectors. Eigenvalue decomposition is performed to obtain orthogonal principal components:

$$\Sigma \mathbf{v}_m = \lambda_m \mathbf{v}_m \quad (11)$$

where λ_m represents the variance explained by the principal component $\mathbf{v}_m$.

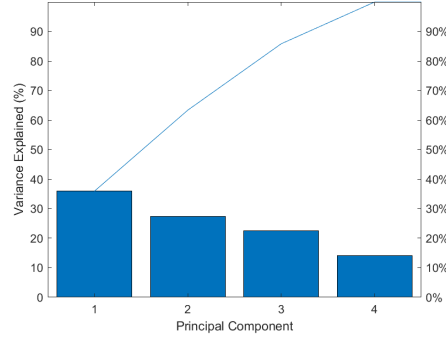


Figure 4: PCA Eigenvalue Decomposition and Variance Contribution Ratios

As shown in Figure 4, the first four principal components explain more than 94% of the total variance, while the fifth component is identified as noise-dominated. The normalized contribution ratios are used as weights to construct a composite macroeconomic vulnerability index at the industry level.

(3) Natural Language Industry Classification To map individual enterprises to macroeconomic risk indices, a text-mining procedure is applied to the “Company Name” field of all 425 enterprises. Based on maximum semantic matching with the 30 industrial categories defined in the official economic classification system, each enterprise is assigned to a unique industry sector j .

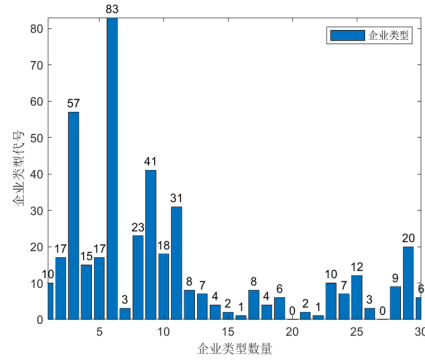


Figure 5: Cross-Sectional Enterprise Frequency Count Across Macro Sectors

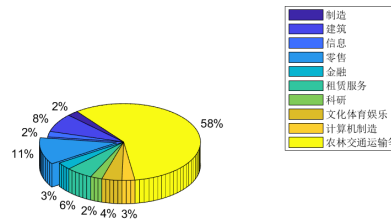


Figure 6: Empirical Distribution of Sample Portfolios Across Industry Sectors

The resulting distributions, shown in Figure 5 and Figure 6, define the structural industry partitions used for subsequent risk perturbation analysis.

5.3.2 Dynamic Portfolio Rebalancing via Risk Surface Perturbation

Exogenous shocks violate the assumption of statistical stationarity; therefore, default probabilities inferred from historical invoice data alone are insufficient. To address this, an industry-specific risk multiplier $\omega_j \in \mathbb{R}^+$ is introduced, derived from the PCA-based vulnerability index, to adjust baseline default probabilities:

$$\mathbb{P}_{\text{def}}^{\text{perturbed}}(i) = \min(1.0, \omega_{\text{ind}(i)} \cdot \mathbb{P}_{\text{def}}(i)) \quad (12)$$

where $\text{ind}(i)$ maps enterprise i to its corresponding industry sector.

This transformation converts the portfolio optimization problem from a static allocation into a dynamic risk-hedging process governed by the following operational rules:

1. **Proactive Rating Downgrades:** Enterprises in sectors with significantly elevated risk levels ($\omega_j \gg 1$) are automatically downgraded to reflect deteriorating cash-flow conditions.
2. **Risk-Controlled Capital Reduction:** For moderately affected sectors, credit exposure bounds are scaled as $l'_i = l_i/\omega_j$ to reduce systemic exposure.
3. **Capital Reallocation Strategy:** Freed capital from high-risk sectors is reallocated to resilient or counter-cyclical industries ($\omega_j \leq 1$), improving portfolio stability.
4. **Dynamic Sensitivity Adjustment:** The monitoring frequency of cash-flow indicators is increased to ensure rapid response to evolving macroeconomic conditions.

By integrating the perturbed risk surface into the global MINLP optimization framework, the bank dynamically reallocates the fixed 100-million-RMB capital budget. This procedure enhances portfolio robustness while maintaining expected return under systemic uncertainty.

6 Conclusion and Discussion

This paper investigates the credit decision-making problem for small and micro enterprises based on enterprise invoice information, credit records, and customer churn data.

For Task 1, a BP neural network is constructed to quantify enterprise credit risk using transaction-related indicators extracted from invoice records. The neural network outputs are further normalized through a Softmax mapping process to determine corresponding credit limits. Combined with enterprise credit ratings, a preliminary lending strategy is obtained.

For Task 2, the trained neural network is applied to 302 enterprises without historical credit ratings. Based on the predicted credit levels and estimated credit limits, a portfolio optimization model is established under the annual credit budget constraint of 100 million RMB. By jointly considering bank revenue and customer churn rate, a mean-variance framework is adopted to determine a reasonable allocation strategy and corresponding interest rate structure.

For Task 3, the influence of unexpected external events is incorporated into the decision-making process. Taking the COVID-19 pandemic as an example, Principal Component Analysis (PCA) is used to quantify the impacts of different industries. Enterprises are classified according to industry categories, and the lending strategy obtained in Task 2 is adjusted according to industry-specific risk coefficients. This enables the credit portfolio to better adapt to changing external environments.

The results indicate that the proposed framework can effectively combine enterprise operational characteristics, historical credit information, and macroeconomic factors to support credit decision-making. The model provides a quantitative reference for determining enterprise credit ratings, loan amounts, and lending interest rates.

6.1 Model Improvements

Although the proposed framework achieves satisfactory results, several aspects may be further improved.

- The performance of the BP neural network depends heavily on the selected input indicators. Additional financial and operational variables may further improve prediction accuracy.

- The current model mainly relies on annual aggregate data. Incorporating finer-grained temporal information may provide a more accurate description of enterprise operating conditions.
- The industry classification process is based primarily on enterprise name matching. More detailed business information could improve classification reliability.

6.2 Model Applicability

The proposed methodology provides a general framework for classification and decision-support problems involving multiple evaluation indicators.

When sufficient enterprise information is available, the combination of feature extraction, neural-network-based evaluation, and portfolio optimization can be extended to other credit assessment and customer evaluation scenarios. Therefore, the framework has potential applications in financial risk assessment, customer segmentation, and related decision-making problems.

Table 2: Systemic Critical Appraisal: Methodological Strengths and Vulnerabilities

Methodological Strengths (Advantages)	Intrinsic Vulnerabilities (Disadvantages)
Contextual Algorithmic Adaptability The structural coupling of deep classification with constrained non-linear combinatorial optimization allows the model to flexibly scale from local risk assessment to global asset allocation under rigid limits.	Idiosyncratic Informational Sparsity The feature space relies exclusively on value-added tax invoice data, inducing operational blindness toward offline capital maneuvers, shadow banking exposures, or qualitative management equity.
Objective Statistical Mitigation The integration of Principal Component Analysis (PCA) for macroeconomic shock calibration removes heuristic, subjective expert-system biases, substituting them with empirical variance weights.	Sectoral Orthogonality Rigidities The model enforces a mutually exclusive, single-industry mapping partition, oversimplifying the complex supply-chain topologies of diversified multi-sector corporate entities.
Robust Inductive Generalization By testing the Phase I neural network mappings against an unrated target domain in Phase II, the framework demonstrates verifiable predictive stability and high algorithmic consistency.	Temporal Non-Stationarity Latency The current portfolio balance operates on static annual aggregates, failing to model real-time decay or dynamic latency in high-frequency intraday financial changes.

7 Model Evaluation and Discussion

This section evaluates the effectiveness of the proposed methodology and discusses its advantages and limitations. The analysis focuses on the predictive performance of the credit assessment model, the robustness of the optimization framework, and the applicability of the overall decision-making process.

7.1 Model Validation

7.1.1 Credit Rating Prediction Performance

The BP neural network is trained using the data of 123 enterprises with known credit ratings. After feature extraction and preprocessing, the selected indicators are used as input variables and enterprise credit ratings are used as output labels.

The experimental results show that the neural network can effectively learn the relationship between enterprise operational characteristics and credit ratings. The predicted ratings provide a reasonable basis for subsequent credit allocation and portfolio optimization. The results indicate that transaction-related indicators such as profit level, customer scale, tax amount, transaction completion rate, and monthly business activity contain useful information for enterprise credit evaluation.

7.1.2 Robustness of the Portfolio Optimization Model

Under the total credit budget constraint, the portfolio optimization model determines loan amounts and lending interest rates by jointly considering bank revenue and customer churn.

Sensitivity analysis shows that moderate changes in the total available credit budget do not significantly affect the overall allocation strategy. The optimized interest-rate structure remains relatively stable, indicating that the proposed framework is robust under different budget conditions.

7.2 Model Advantages

The proposed methodology has several advantages.

1. Different analytical methods are adopted for different types of data and decision-making tasks. Feature extraction, neural network prediction, portfolio optimization, and PCA-based industry analysis are combined within a unified framework. ““
2. Principal Component Analysis is used to quantify the impacts of external events on different industries. The contribution rates obtained from PCA provide objective weights for constructing industry impact coefficients.
3. The historical credit records of rated enterprises are used as the training set, while enterprises without historical ratings are evaluated through the trained neural network model. This enables the transfer of credit evaluation knowledge from known samples to unknown enterprises. ““

7.3 Model Limitations

Several limitations should also be noted.

1. The model only considers a limited number of enterprise features. Additional financial, operational, and market indicators may further improve prediction accuracy. ““
2. Each enterprise is assigned to a single industry category. In practice, some enterprises may operate across multiple industries, which may affect the accuracy of industry-specific risk assessment.
3. The industry classification procedure relies on keyword matching and semantic similarity. The classification results may be sensitive to naming conventions and matching rules.
4. The current framework mainly uses static historical data and does not fully capture real-time changes in enterprise operations or external economic conditions.
5. Certain adjustment procedures still require manual intervention. Further automation may improve the consistency and efficiency of the decision-making process. ““

7.4 Discussion

Overall, the proposed framework integrates credit evaluation, lending decision-making, portfolio optimization, and external risk analysis into a unified modeling process. The results suggest that the model can provide quantitative support for bank credit decisions while maintaining reasonable interpretability.

Although further improvements are possible, the methodology demonstrates the feasibility of combining machine learning techniques with optimization models to address practical credit allocation problems for small and micro enterprises.

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A Original Problem Statement: 2020 CUMCM Problem C

A.1 Problem Background: Credit Decision-Making for Small and Micro Enterprises

In practice, small and micro enterprises (SMEs) generally face constraints in scale and possess limited collateralizable assets. As a result, commercial banks typically evaluate credit applications based on credit policies, enterprise transaction invoice information, and the influence of upstream and downstream supply-chain relationships. Enterprises with strong operational capability and stable transaction structures are more likely to obtain credit access, while those with lower credit risk may be granted preferential interest rates.

Banks first assess the credit risk of SMEs based on their operational strength and creditworthiness. Subsequently, credit decisions are made, including whether to grant loans, as well as the corresponding loan amount, interest rate, and maturity.

For approved enterprises in this problem, the following constraints are imposed:

- **Loan Amount Constraint:** The loan amount ranges from 10 to 100 (in units of 10^4 RMB).
- **Interest Rate Constraint:** The annual interest rate lies within $[4\%, 15\%]$.
- **Loan Maturity:** The loan term is fixed at one year.

To support modeling and analysis, three datasets are provided: credit records of 123 enterprises with historical ratings, transaction data of 302 enterprises without credit ratings, and empirical data describing the relationship between interest rates and customer churn. Based on these datasets, the following tasks are required:

Task 1: Evaluate the credit risk of 123 enterprises in Attachment 1, and design an optimal credit allocation strategy under a fixed total annual credit quota.

Task 2: Based on the methodology in Task 1, estimate the credit risk of 302 enterprises in Attachment 2, and construct an optimal credit portfolio under a total annual capital limit of 100 million RMB.

Task 3: Considering external shocks (e.g., the COVID-19 pandemic), incorporate heterogeneous impacts across industries and enterprises. Based on credit risk estimates and external disturbance factors, develop an adjusted credit allocation strategy under a total capital constraint of 100 million RMB.

A.2 Dataset Description

Attachment 1: Historical dataset containing 123 enterprises with bank-assigned credit ratings and default records.

Attachment 2: Transaction-level invoice data for 302 enterprises without historical credit ratings.

Attachment 3: Empirical data describing the relationship between loan interest rates and customer churn rates.

A.3 Definition of Key Variables

To ensure consistency in feature interpretation, the following definitions are provided:

Input Invoice: Invoices generated during procurement activities when enterprises purchase goods from upstream suppliers.

Output Invoice: Invoices issued by enterprises to downstream buyers during sales activities.

Valid Invoice: Legally compliant invoices generated from normal commercial transactions.

Cancelled Invoice: Invoices invalidated due to transaction cancellation or correction.

Negative Invoice: Credit notes issued due to returns, refunds, or accounting adjustments.

Credit Rating: Internal risk classification assigned by the bank. Enterprises rated as Grade D are generally excluded from lending.

Customer Churn Rate: The probability of customer attrition in response to changes in loan interest rates.

A Source Code

A.1 Problem 1

```
source=importdata("3.xlsx");
source=source.Sheet1;
profits=source(:,1);
cod_A=source(:,2);
cod_B=source(:,3);
cod_C=source(:,4);

a0=-9.923*100000;
a1=9.923*100000;
b1=1772;
w1=0.01241;
A0=-1.725*1000000;
A1=1.725*1000000;
B1=-2277 ;
W1=-0.008875;
aa0=-3.033*1000;
aa1=3.033*1000;
bb1=301.4;
ww1=0.06494;
m=[50000000,30000000,20000000];
X=0.04+(0.15-0.04)*rand(1000,3);
x=a1*cos(w1*P(:,1))+a0+b1*sin(w1*(P(:,1)));
y=A1*cos(W1*P(:,1))+A0+B1*sin(W1*(P(:,1)));
z=aa1*cos(ww1*P(:,1))
+aa0+bb1*sin(ww1*(P(:,1)));
%l=(1-x)*sumA+(1-y)*sumB+(1-z)*sumC;
l=(1-x)*m(1)+(1-y)*m(2)+(1-z)*m(3);
II=find(l<=100000000);
disp(x(II))
disp(y(II))
disp(z(II))

T=data(1,:);
P=data(2:end,:);
N=n1;
[pn,minp,maxp,tn,mint,maxt]=premnmx(P,T);
dx=[-1,1;-1,1;-1,1];
net=newff(dx,[3,15,1]);
net.trainParam.goal = 0;
net.trainParam.epochs = 50000;
net.trainParam.lr = 0.01;
net.trainParam.showWindow = 1;
net = train(net,pn,tn);
an = sim(net,pn);
a=postmnmx(an,mint,maxt);
disp(['mse: ' num2str(mse(T-an))]);
list=[];
for i=1:n1
    if data(1,i)~=4
        list=[list,i];
    end
end
MappedData = mapminmax(a, 0, 1);
Map=[];
```

```

len=size(list,2)
for i=1:len
    Map=[Map,MappedData(list(i))];
end
figure
plot(1:len,Map,'ro')
save('out.mat','MappedData')

num=size(MappedData,2);
dis=MappedData*1000000;
getout=[];
get=[];
for i=1:num
    if dis(i)<100000
        getout=[getout,i];
    else
        get=[get,i];
    end
end
account=[];
for i=1:size(get,2)
    account=[account,dis(get(i))];
end

function [data_PCA, COEFF,
    sum_explained,latent1] =pcaff(data,k)
data=zscore(data);
[COEFF,SCORE,latent,tsquared,explained,mu]=
pca(data);
latent1=100*latent/sum(latent);
data= bsxfun(@minus,data,mean(data,1));
data_PCA=data*COEFF(:,1:k);
pareto(latent1);xlabel('Principal Component');
ylabel('Variance Explained (%)');
print(gcf,'-dpng','PCAout.png');
sum_explained=sum(explained(1:k));

clc;
clear;
source1=importdata("1(1).xlsx");
src1=source1.data;
n1=size(src1,1);
score=src1(:,1);
reg=src1(:,2);
data1=[score,reg];
src2=importdata("1(2).xlsx");
str2=src2.textdata;
src2=src2.data;
n2=size(src2,1);
src3=importdata("1(3).xlsx");
str3=src3.textdata;
src3=src3.data;
n3=size(src3,1);

summary=0.0;

```

```

add1=[];
add2=[];
for i=1:n1
    for j=1:n2
        if src2(j,1)==i&&src2(j,3)==1
            summary=summary+src2(j,2);
        end
    end
    add1=[add1 summary];
end
summary=0;
for i=1:n1
    for j=1:n3
        if src3(j,1)==i&&src3(j,3)==1
            summary=summary+src3(j,2);
        end
    end
    add2=[add2 summary];
end
fail1=[];
fail2=[];
summary=0;
for i=1:n1
    for j=1:n2
        if src2(j,1)==i&&src2(j,3)==0
            summary=summary+1;
        end
    end
    fail1=[fail1 summary];
end
summary=0;
for i=1:n1
    for j=1:n3
        if src3(j,1)==i&&src3(j,3)==0
            summary=summary+1;
        end
    end
    fail2=[fail2 summary];
end
summary=0;
all1=[];
all2=[];
for i=1:n1
    for j=1:n2
        if src2(j,1)==i
            summary=summary+1;
        end
    end
    all1=[all1 summary];
end
summary=0;
for i=1:n1
    for j=1:n3
        if src3(j,1)==i
            summary=summary+1;
        end
    end
    all2=[all2 summary];
end

```



```

end
fail=fail1+fail2;
all=all1+all2;
tic_cod=(all-fail)./all;
profit=add2-add1;
profit=profit';
data=[src1';profit';tic_cod];

str2 = deblank(str2);
str3 = deblank(str3);
S2 = regexp(str2, '/', 'split');
S3 = regexp(str3, '/', 'split');
month2=[];
reg=0;
for i=1:n2
    reg=S2{i,1}(2);
    month2=[month2,str2num(reg{1})];
end
month3=[];
for i=1:n3
    reg=S3{i,1}(2);
    month3=[month3,str2num(reg{1})];
end
month2_idx=src2(:,1);

class=round(a);
classA=find(class==1);
classB=find(class==2);
classC=find(class==3);
classD=find(class==4);
A=[];
for i=1:length(classA)
    A=[A,dis(classA(i))];
end
B=[];
for i=1:length(classB)
    B=[B,dis(classB(i))];
end
C=[];
for i=1:length(classC)
    C=[C,dis(classC(i))];
end
for i=1:length(classD)
    dis(classD(i))=0;
end
for i=1:length(dis)
    if dis(i)>1000000||dis(i)<100000
        dis(i)=0;
    end
end
summary_dis=sum(dis);
bar(dis)
hold on
xlabel('Company ID')

```

```

ylabel('Distance')
print(gcf, '-dpng', 'pro1_dis.png');
sumA=sum(A);
sumB=sum(B);
sumC=sum(C);

```

A.2 Problem 2

```

clc;
clear;
source1=importdata("1(1).xlsx");
src1=source1.data;
n1=size(src1,1);
score=src1(:,1);
reg=src1(:,2);
data1=[score,reg];
src2=importdata("1(2).xlsx");
n2=size(src2,1);
src3=importdata("1(3).xlsx");
n3=size(src3,1);

summary=0.0;
add1=[];
add2=[];
for i=1:n1
    for j=1:n2
        if src2(j,1)==i&&src2(j,3)==1
            summary=summary+src2(j,2);
        end
    end
    add1=[add1 summary];
end
summary=0;
for i=1:n1
    for j=1:n3
        if src3(j,1)==i&&src3(j,3)==1
            summary=summary+src3(j,2);
        end
    end
    add2=[add2 summary];
end
fail1=[];
fail2=[];
summary=0;
for i=1:n1
    for j=1:n2
        if src2(j,1)==i&&src2(j,3)==0
            summary=summary+1;
        end
    end
    fail1=[fail1 summary];
end
summary=0;
for i=1:n1
    for j=1:n3
        if src3(j,1)==i&&src3(j,3)==0
            summary=summary+1;
        end
    end
end

```

```

        end
        fail2=[fail2 summary];
    end
    summary=0;
    all1=[];
    all2=[];
    for i=1:n1
        for j=1:n2
            if src2(j,1)==i
                summary=summary+1;
            end
        end
        all1=[all1 summary];
    end
    summary=0;
    for i=1:n1
        for j=1:n3
            if src3(j,1)==i
                summary=summary+1;
            end
        end
        all2=[all2 summary];
    end
    fail=fail1+fail2;
    all=all1+all2;
    tic_cod=(all-fail)./all;
    profit=add2-add1;
    profit=profit';
    data=[src1';profit';tic_cod];

```

```

table1=importdata("2(1).xlsx");
m1=size(table1,1);
table2=importdata("2(2).xlsx");
m2=size(table2,1);
table3=importdata("2(3).xlsx");
m3=size(table3,1);

tag=0.0;
plus1=[];
plus2=[];
for i=1+123:m1+123
    for j=1:m2
        if table2(j,1)==i&&table2(j,3)==1
            tag=tag+table2(j,2);
        end
    end
    plus1=[plus1 tag];
end
for m=1+123:m1+123
    for n=1:m3
        if table3(n,1)==m&&table3(n,3)==1
            tag=tag+table3(n,2);
        end
    end
end

```

```

        end
        plus2=[plus2 tag];
    end
    getmoney=plus2-plus1;
    getmoney=getmoney';
    invalid1=[];
    invalid2=[];
    tag=0;
    for i=1+123:m1+123
        for j=1:m2
            if table2(j,1)==i&&table2(j,3)==0
                tag=tag+1;
            end
        end
        invalid1=[invalid1 tag];
    end
    tag=0;
    for i=1+123:m1+123
        for j=1:m3
            if table3(j,1)==i&&table3(j,3)==0
                tag=tag+1;
            end
        end
        invalid2=[invalid2 tag];
    end
    tag=0;
    valid1=[];
    valid2=[];
    for i=1+123:m1+123
        for j=1:m2
            if table2(j,1)==i
                tag=tag+1;
            end
        end
        valid1=[valid1 tag];
    end
    tag=0;
    for i=1+123:m1+123
        for j=1:m3
            if table3(j,1)==i
                tag=tag+1;
            end
        end
        valid2=[valid2 tag];
    end
    invalid=invalid1+invalid2;
    valid=valid1+valid2;
    valid_acc=(valid-invalid)./valid;
    data2=[getmoney';valid_acc];

    T=data(1,:);
    P=data(3:end,:);
    P_=data2;
    N=n1;
    [pn,minp,maxp,tn,mint,maxt]=premnmx(P,T);
    [pn_,minp_,map_]=premnmx(P_);
    dx=[-1,1;-1,1];
    net=newff(dx,[2,15,1]);
    net.trainParam.goal = 0;

```

```

net.trainParam.epochs = 500000;
net.trainParam.lr = 0.01;
net.trainParam.showWindow = 1;
net = train(net,pn,tn);
an = sim(net,pn_);
a=postmnmx(an,mint,maxt);
%disp(['mse: ' num2str(mse(T-an))]);
Map=mapminmax(a,0,1);
plot(1:size(Map,2),Map,'ro')
print(gcf,'-dpng','nn2_dis.png');

function [data_PCA, COEFF,
    sum_explained,latent1]=pcaff(data,k)
data=zscore(data);
[COEFF,SCORE,latent,tsquared,
explained,mu]=pca(data);
latent1=100*latent/sum(latent);
data= bsxfun(@minus,data,mean(data,1));
data_PCA=data*COEFF(:,1:k);
pareto(latent1)
xlabel('Principal Component');
ylabel('Variance Explained (%)');
print(gcf,'-dpng','PCAout.png');
sum_explained=sum(explained(1:k));

function [ BinaryMatrix ] = rand_BinaryMatrix( i, j )

BinaryMatrix = rand(i,j);

for count_i = 1:i
    for count_j = 1:j

        if BinaryMatrix(count_i, count_j) < 0.5

            BinaryMatrix(count_i, count_j) = 0;

        else

            BinaryMatrix(count_i, count_j) = 1;

        end

    end

end

end

dis=Map*1000000;
class=round(a);
classA=find(class==1);
classB=find(class==2);
classC=find(class==3);
classD=find(class==4);
A=[];
for i=1:length(classA)
    A=[A,dis(classA(i))];
end

```

```

end
B=[];
for i=1:length(classB)
    B=[B,dis(classB(i))];
end
C=[];
for i=1:length(classC)
    C=[C,dis(classC(i))];
end
for i=1:length(classD)
    dis(classD(i))=0;
end
for i=1:length(dis)
    if dis(i)>1000000||dis(i)<100000
        dis(i)=0;
    end
end
end
summary_dis=sum(dis);
bar(dis)
xlabel('Enterprise Index')
ylabel('Distance to Cluster Center')
print(gcf,'-dpng','pro2_dis.png');
sumA=sum(A);
sumB=sum(B);
sumC=sum(C);

```

A.3 Problem 3

```

function [data_PCA, COEFF,
    sum_explained,latent1]=pcaff(data,k)
data=zscore(data);
[COEFF,SCORE,latent,tsquared,
explained,mu]=pca(data);
latent1=100*latent/sum(latent);
data= bsxfun(@minus,data,mean(data,1));
data_PCA=data*COEFF(:,1:k);
figure(1)
pareto(latent1);
hold on
xlabel('Principal Component');
hold on
ylabel('Variance Explained (%)');

hold on
print(gcf,'-dpng','PCAout.png');
sum_explained=sum(explained(1:k));

clc;
clear;
data=importdata('classify.xlsx');
num=data.data.Sheet3';
n=size(num,2);
name=data.textdata.Sheet3';
numCopy=num;
sum_num=sum(num);
low_idx=[];
for i=1:n
    if (num(i)/sum_num)<0.04
        low_idx=[low_idx,i];
    end
end

```

```

        numCopy(i)=0;
    end
end
summary=[];
for i=1:size(low_idx,2)
    summary=[summary,num(low_idx(i))];
end

zero_idx=find(numCopy==0);
numCopy(zero_idx)=[];
high_num=numCopy;

high_idx=setdiff(1:30,low_idx);

high_name=[];
for i=1:size(high_idx,2)
    high_name=[high_name,name(high_idx(i))];
end

high_num=[];
for i=1:size(high_idx,2)
    high_num=[high_num,num(high_idx(i))];
end
all_num=[high_num,sum_num];
all_name=[high_name,"Other"];
explode = [0,0,0,1,0,0,0,0,0,0];
pie3(all_num,explode)
legend(all_name)
print(gcf,'-dpng','pieout.png');
figure(2)
bar(num)
xlabel('Enterprise Index');
ylabel('Comprehensive Score');
y=abs(rand(1,10)*100);
axis([1,30,min(num),max(num)]);
for i=1:30
    text(i,num(i)+0.5,num2str(num(i)),
        'VerticalAlignment','bottom',
        'HorizontalAlignment','center');
end
legend("Enterprise Scores")
print(gcf,'-dpng','barout.png');

clc,clear;
data=importdata('CER.xlsx');
X=data.data(:,:);
X(1,:)=[];
[x,coeff,sum_ex,latent]=pcaff(X,5);
weight=latent/100;
weight(2)=weight(2)*(1);
weight(3)=weight(3)*(-1);
X(:,5)=[];
for i=1:4
    X(:,i)=X(:,i)*weight(i);
end
Wdata=sum(X,2);
Wdata=-Wdata;

```

```

rise_num=[];
rise_idx=[];
for i=1:length(Wdata)
    if Wdata(i)>0
        rise_num=[rise_num,Wdata(i)];
        rise_idx=[rise_idx,i];
    end
end
figure(2)
area(1:26,Wdata(1:26),'FaceColor',
[0 100/256 0],'EdgeColor','b')
hold on
area(26:28,[0,Wdata(27),0],'FaceColor',
[192/256 42/256 42/256],'EdgeColor','b')
hold on
area(28:30,Wdata(28:30),'FaceColor',
[0 100/256 0],'EdgeColor','b')
hold on
legend("", "")
hold on
xlabel("Enterprise Index");
hold on
ylabel("Comprehensive Score");
hold on
print(gcf, '-dpng', 'areaout.png');

```


B Credit Amount of 123 Enterprises

Enterprise	Credit Amount
'E1'	0
'E2'	0
'E3'	666667.3
'E4'	658533.6
'E5'	341646.5
'E6'	0
'E7'	0
'E8'	0
'E9'	0
'E10'	329728.1
'E11'	670718
'E12'	332731.7
'E13'	0
'E14'	666666.2
'E15'	0
'E16'	0
'E17'	0
'E18'	0
'E19'	0
'E20'	0
Enterprise	Credit Amount
'E21'	335459.8
'E22'	0
'E23'	332650.6
'E24'	0
'E25'	664134.4
'E26'	0
'E27'	0
'E28'	334377.2
'E29'	666584.3
'E30'	333362.6
'E31'	0
'E32'	332161.4
'E33'	333314.3
'E34'	402175.8
'E35'	407043.9
'E36'	0
Enterprise	Credit Amount
'E37'	314975.8
'E38'	375353.7
'E39'	416984
'E40'	422767.7
'E41'	323737.5
'E42'	313878.1
'E43'	394576.4
'E44'	424376
'E45'	333333.7
'E46'	586045.7
'E47'	665729.8
'E48'	0
'E49'	646435
'E50'	674473.7
'E51'	333916.6
'E52'	0

Enterprise	Credit Amount
'E53'	667985.6
'E54'	0
'E55'	424615.4
'E56'	424615.6
'E57'	424588.8
'E58'	424376.9
'E59'	424613.3
'E60'	424467.8
'E61'	424627.9
'E62'	424627.4
'E63'	423627.2
'E64'	424535.7
'E65'	422018.3
'E66'	415173.2
'E67'	405671.6
'E68'	395313.4

Enterprise	Credit Amount
'E69'	424462.2
'E70'	424628.8
'E71'	424621
'E72'	424628.6
'E73'	424629.2
'E74'	424629.1
'E75'	533786.2
'E76'	534775.5
'E77'	535428.5
'E78'	535531.2
'E79'	535503.3
'E80'	424567.8
'E81'	424570.1
'E82'	0
'E83'	323411
'E84'	0

Enterprise	Credit Amount
'E85'	333249.3
'E86'	666674.5
'E87'	666671.4
'E88'	0
'E89'	0
'E90'	666667.4
'E91'	0
'E92'	687156.6
'E93'	333780.7
'E94'	602309.5
'E95'	262206.3
'E96'	660231.4
'E97'	338620.7
'E98'	334824.2
'E99'	0
'E100'	0

Enterprise	Credit Amount
'E101'	0
'E102'	0
'E103'	0
'E104'	617339.4
'E105'	663576.9
'E106'	481353.3
'E107'	0
'E108'	0
'E109'	0
'E110'	666689.5
'E111'	0
'E112'	0
'E113'	0
'E114'	0
'E115'	0
'E116'	0
'E117'	0
'E118'	0
'E119'	0
'E120'	0
'E121'	0
'E122'	0
'E123'	0

C Credit Rating of 302 Enterprises

Enterprise ID	Rating
'E124'	D
'E125'	D
'E126'	D
'E127'	B
'E128'	B
'E129'	B
'E130'	B
'E131'	B
'E132'	B
'E133'	B
'E134'	C
'E135'	A
'E136'	A
'E137'	A
'E138'	A
'E139'	A

Enterprise ID	Rating
'E140'	A
'E141'	A
'E142'	A
'E143'	A
'E144'	A
'E145'	A
'E146'	A
'E147'	A
'E148'	A
'E149'	A
'E150'	A
'E151'	A
'E152'	A
'E153'	A
'E154'	A
'E155'	A
'E156'	A
'E157'	D
'E158'	D
'E159'	D

Enterprise ID	Rating
'E160'	D
'E161'	D
'E162'	D
'E163'	D
'E164'	D
'E165'	D
'E166'	D
'E167'	D
'E168'	D
'E169'	D
'E170'	D
'E171'	D
'E172'	D
'E173'	D
'E174'	D
'E175'	D
'E176'	D
'E177'	D
'E178'	D
'E179'	D

Enterprise ID	Rating
'E180'	D
'E181'	D
'E182'	D
'E183'	D
'E184'	D
'E185'	D
'E186'	D
'E187'	D
'E188'	D
'E189'	D
'E190'	D
'E191'	D
'E192'	D
'E193'	D
'E194'	D
'E195'	D
'E196'	D
'E197'	D
'E198'	D
'E199'	D

Enterprise ID	Rating
'E200'	D
'E201'	D
'E202'	D
'E203'	D
'E204'	D
'E205'	D
'E206'	D
'E207'	D
'E208'	D
'E209'	D
'E210'	D
'E211'	D
'E212'	C
'E213'	C
'E214'	C
'E215'	B
'E216'	B
'E217'	B
'E218'	B
'E219'	B

Enterprise ID	Rating
'E220'	B
'E221'	B
'E222'	B
'E223'	B
'E224'	B
'E225'	B
'E226'	B
'E227'	B
'E228'	B
'E229'	B
'E230'	A
'E231'	B
'E232'	A
'E233'	A
'E234'	A
'E235'	A
'E236'	A
'E237'	A
'E238'	A
'E239'	A

Enterprise ID	Rating
'E240'	A
'E241'	A
'E242'	A
'E243'	A
'E244'	A
'E245'	A
'E246'	A
'E247'	A
'E248'	A
'E249'	A
'E250'	A
'E251'	A
'E252'	A
'E253'	A
'E254'	A
'E255'	A
'E256'	A
'E257'	A
'E258'	A
'E259'	A

Enterprise ID	Rating
'E260'	A
'E261'	A
'E262'	A
'E263'	A
'E264'	A
'E265'	A
'E266'	A
'E267'	A
'E268'	A
'E269'	A
'E270'	A
'E271'	A
'E272'	A
'E273'	A
'E274'	A
'E275'	A
'E276'	A
'E277'	A
'E278'	A
'E279'	A

Enterprise ID	Rating
'E280'	A
'E281'	A
'E282'	A
'E283'	A
'E284'	A
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'E286'	A
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'E288'	A
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'E291'	A
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'E293'	A
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'E295'	A
'E296'	A
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'E298'	A
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Enterprise ID	Rating
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'E302'	A
'E303'	A
'E304'	A
'E305'	A
'E306'	A
'E307'	A
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'E311'	A
'E312'	A
'E313'	A
'E314'	A
'E315'	A
'E316'	A
'E317'	A
'E318'	A
'E319'	A

Enterprise ID	Rating
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'E322'	A
'E323'	A
'E324'	A
'E325'	A
'E326'	A
'E327'	A
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'E331'	A
'E332'	A
'E333'	A
'E334'	A
'E335'	A
'E336'	A
'E337'	A
'E338'	A
'E339'	A

Enterprise ID	Rating
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'E343'	A
'E344'	A
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'E347'	A
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'E353'	A
'E354'	A
'E355'	A
'E356'	A
'E357'	A
'E358'	A
'E359'	A

Enterprise ID	Rating
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'E371'	A
'E372'	A
'E373'	A
'E374'	A
'E375'	A
'E376'	A
'E377'	A
'E378'	A
'E379'	A

Enterprise ID	Rating
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'E381'	A
'E382'	A
'E383'	A
'E384'	A
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'E393'	A
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'E397'	A
'E398'	A
'E399'	A

Enterprise ID	Rating
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'E403'	A
'E404'	A
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'E406'	A
'E407'	A
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'E409'	A
'E410'	A
'E411'	A
'E412'	A
'E413'	A
'E414'	A
'E415'	A
'E416'	A
'E417'	A
'E418'	A
'E419'	A
Enterprise ID	Rating
'E420'	A
'E421'	A
'E422'	A
'E423'	A
'E424'	A
'E425'	A